CS758-Lecture02

Topics

- Basic definitions and representation
- Gram-Schmidt algorithm
- Characterising lattice bases

Sources

- [1]: BOOK-NguVal10 STUB, Chapter 02 (PDF)
- [2]: COURSE-Micciancio07, Lecture 02 (PDF)

Plan

- At the end of the lecture, you will have understood:
  - The difference between vector (sub)spaces and lattices
  - Basic properties of lattice bases
- Basic definitions and representation
  - We work over an n-dimensional vector space (to be specific, the Euclidean space) equipped with the standard notion of inner product (and thus norm, i.e., $\|\bar{x}\| := \langle \bar{x}, \bar{x} \rangle = \sum_{i=1}^{n} x_i^2$)
  - "Topological" definition via discreteness
    - Topologically, a lattice is a set of "regular" "well-spaced" points in $\mathbb{R}^n$
    - **Definition 1** (see [1], Definition 1): A subset $\mathcal{S}$ of $\mathbb{R}^n$ is called discrete if for every $\bar{x} \in \mathcal{S}$, there exists a $\rho > 0$ such that

      $$B(\bar{x}, \rho) \cap \mathcal{S} = \{\bar{x}\},$$

      where $B(\bar{x}, \rho)$ is the open ball of radius ρ centred at $\bar{x}$
    - **Definition 2** (see [1], Definition 4): A lattice Λ of $\mathbb{R}^n$ is a *discrete subgroup* of $(\mathbb{R}^n, +)$. Recall that a subgroup of $(\mathbb{R}^n, +)$ is a subset of $\mathbb{R}^n$ that is also a group.
    - **PQ1**. Lattice or not?
      1. $\{0^n\}$: Yes. It is the trivial "zero" lattice
      2. $\mathbb{N}^n$: No. Although it is discrete (e.g., for $\rho = 0.5$), it is not a subgroup of $\mathbb{R}^n$
      3. $\mathbb{Z}^n$: Yes. It is the integer lattice
      4. $\mathbb{Q}^n$: No. Although it is a subgroup of $\mathbb{R}^n$, it is not discrete (HW)
      5. $\{(x_1, \cdots, x_n) \in \mathbb{Z}^n : \sum_{i=1}^{n} a_i x_i = 0 \bmod q\}$, where $a_1, \cdots, a_n, q \in \mathbb{N}$. Yes. It is discrete since any subset of a discrete set is discrete. It can also be shown to be a subgroup (HW).
    - **HW1**:

1. What happens if we flip the order of quantifiers in Definition 1? That is, require that there exists a $\rho > 0$ for every $\bar{x} \in \mathcal{S}$, such that

$$B(\bar{x}, \rho) \cap \mathcal{S} = \{\bar{x}\}?$$

2. In Definition 2, what if you require Λ to be a discrete *subspace*?
3. Can the cardinality of Λ be *uncountable*?

- "Algebraic/geometric" definition via basis
  - Algebraically/geometrically, a lattice is the set of all vectors in the "integer span" of a basic $\bar{B}$
  - **Definition 3**. Let $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m)$ be *linearly independent* vectors in $\mathbb{R}^n$. The lattice *generated by* $\bar{B}$ is defined as

$$\Lambda(\bar{B}) := \{\bar{B}\bar{x} : \bar{x} \in \mathbb{Z}^m\} = \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : x_1, \cdots, x_m \in \mathbb{Z} \right\}$$

    - **Example 1.** Consider the lattice generated by $\bar{B} := ((3,0)^t, (1,2)^t)$
    - *We* will refer to n as the *dimension* of Λ and m as its *rank*. The lattice is full rank if $m = n$.
      - Note: this terminology is not consistent across papers (e.g., Regev's lecture notes follows this)
  - **HW2.** (Every lattice according to Definition 1 has at least one basis): Let Λ be a lattice as per Definition 1. For some $m \leq n$, show that there exists m linearly independent vectors $\bar{b}_1, \cdots, \bar{b}_m \in \mathbb{R}^n$ such that Λ is generated by $\bar{b}_1, \cdots, \bar{b}_m$.

- Lattice *vs* vector space ([2])
  1. A basis for a lattice is also a basis for the vector space, but converse is not true:
     - **Example 2.** Given a basis $\bar{B}$ (e.g., $(1,0)^t$ and $(0,1)^t$) for a lattice Λ (e.g., $\mathbb{Z}^2$), $2\bar{B}$ is still a basis for the vector space, but not Λ.
  2. Vector spaces *always* admit an *orthogonal* basis and these can be efficiently computed using the Gram-Schmidt algorithm:
     - **Algorithm 1**:
       - *Input*: Basis $\bar{b}_1, \cdots, \bar{b}_m \in \mathbb{R}^n$ for some vector space $\mathcal{V}$
       - *Output*: Orthogonal basis $\bar{b}_1^*, \cdots, \bar{b}_m^* \in \mathbb{R}^n$ for $\mathcal{V}$
       - Algorithm:
         1. Set $\bar{b}_1^* := \bar{b}_1$
         2. Compute $\bar{b}_2^* := \bar{b}_2 - \pi_{\bar{b}_1^*}(\bar{b}_2)$, where $\pi_{\bar{u}}(\bar{v})$ denotes the orthogonal projection of $\bar{b}_2$ on to $\bar{b}_1^*$, i.e.,

$$\frac{\langle \bar{b}_2, \bar{b}_1^* \rangle}{\langle \bar{b}_1^*, \bar{b}_1^* \rangle} \bar{b}_1^*$$

         3. For each $i \in [3, m]$: compute $\bar{b}_i^* := \bar{b}_i - \sum_{j<i} \pi_{\bar{b}_j^*}(\bar{b}_i)$ (i.e., pick $\bar{b}_i^*$ to be orthogonal to span of all previous vectors)

4. Output $\bar{b}_1^*, \cdots, \bar{b}_m^*$

- **Example 3.** On running Gram-Schmidt algorithm on $\bar{B} := ((3,0)^t, (1,2)^t)$ results in $((3,0)^t, (0,2)^t)$
- **Fact 1 (Theorem 9 in [2]).** Gram-Schmidt algorithm runs in polynomial time.
- **PQ2.** Does Gram-Schmidt algorithm yield an orthogonal *lattice* basis?
  - Not necessarily, in Example 2: $(0,2)^t$ output above is not even in the lattice $\Lambda(\bar{B})$!
  - Where/why does it fail? The value of $\langle \bar{b}_2, \bar{b}_1^* \rangle / \langle \bar{b}_1^*, \bar{b}_1^* \rangle$ in the above equation need not be an integer.

- Characterising lattice bases
  - In general, lattice bases are *much more complex* than vector space bases! For example, some lattices do not even have an orthogonal basis!
    - **Example 4.** The lattice generated by $(2,0)^t$ and $(1,2)^t$ contains orthogonal vectors (e.g, $(0,4)$ and $(2,0)$), but it doesn't have an orthogonal basis. (HW: argue formally)
  - Looking ahead, this is exactly why lattices are interesting from an algorithmic/cryptographic point of view: several problems that are computationally easy over vector spaces (e.g., finding orthogonal basis) become computationally hard over lattices! To understand lattice bases better, let's start with the following question.
  - **PQ3.** Is the basis for vector space unique? What about lattices?
    - Basis for vector spaces need not be unique. Similarly, same lattice can have more than one basis.
    - **Example 5.**: $\bar{B}_1 := ((2,0)^T, (2,2)^T)$ and $\bar{B}_2 := ((2,0)^T, (0,-2)^T)$ are also basis of the scaled integer lattice $2\mathbb{Z}^n$.
  - Let's try to characterise all the basis of a lattice. This this end we will need two definitions: fundamental parallelepiped and determinant.
    - **Definition 4.** (Fundamental parallelepiped) Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n \times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

$$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : 0 \leq x_i < 1 \right\}$$

      - Geometrically, when $m = n$, the whole space $\mathbb{R}^n$ can be obtained by "tiling" using $P(\bar{B})$.
    - **Definition 5.** (Determinant) Let $\bar{B} \in \mathbb{R}^{n \times m}$ be a basis. The determinant of $\Lambda(\bar{B})$ is the m-dimensional volume of $P(\bar{B})$:

$$det(\Lambda(\bar{B})) = vol(P(\bar{B}))$$

      - Geometrically, it captures density of lattice points in its ambient space: if the determinant is small then the lattice is dense and vice versa

- **HW4 (Hadamard inequality).** Show that $det(\Lambda(\bar{B})) \leq \Pi_{i=1}^{m} ||\bar{b}_i||$
- **HW5 (Determinant is efficiently computable).** Let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*) \in \mathbb{R}^{n \times m}$ be the basis obtained by running the Gram-Schmidt algorithm on $\bar{B}$ in Definition 5. Then

$$det(\Lambda(\bar{B})) = \Pi_{i=1}^{m} ||\bar{b}_i^*||.$$

- **HW6 (Determinant is multiplicative.)** For any $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m \times m}$, show that

$$det(\bar{M}_1 \bar{M}_2) = det(\bar{M}_1) det(\bar{M}_2).$$

- **Theorem 1.** ([2], Theorem 2) Let $\bar{B}_1, \bar{B}_2 \in \mathbb{R}^{n \times m}$ be two bases. Then $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$ iff there exists a unimodular matrix $\bar{U} \in \mathbb{Z}^{m \times m}$ such that $\bar{B}_2 = \bar{B}_1 \bar{U}$, where an (integer) matrix $\bar{U}$ is unimodular if $det(\bar{U}) = \pm 1$.
  - Intuitively, a unimodular matrix preserves "volume" of the fundamental parallelepiped.
  - **PQ4.** Given Theorem 1, how do you enumerate all bases for a lattice Λ?
  - **Proof of Theorem 1.**
    - We first prove the following lemma about unimodular matrices:
      - **Lemma 1.** If $\bar{U} \in \mathbb{Z}^{m \times m}$ is a unimodular matrix, then so is $\bar{U}^{-1}$
      - **Proof.** We split the proof into two parts:
        1. In the first part, we argue that $\bar{U}^{-1}$ is an *integer* matrix. To see why, parse $\bar{U}^{-1}$ as $(\bar{v}_1, \cdots, \bar{v}_m)$ and each $\bar{v}_i$ as $(v_{i,1}, \cdots, v_{i,m})^t$ and consider the equation

$$\bar{U}\bar{U}^{-1} = \bar{I}_m.$$

           Using Cramer's algorithm, the system of linear equations

$$\bar{U}\bar{v}_i = \bar{e}_i,$$

           where $\bar{e}_i$ is the ith unit vector, has a unique solution $\bar{v}_i$ where

$$v_{i,j} := \frac{det((\bar{u}_1, \cdots, \bar{u}_{i-1}, \bar{e}_i, \bar{u}_{i+1}, \cdots, \bar{v}_n))}{det(\bar{U})}.$$

           Since all the vectors in the numerator are integer, and $det(\bar{U}) = \pm 1$, we have that $\bar{v}_i$ is also an integer vector and thus $\bar{U}^{-1}$ is also an integer vector.
        2. In the second part, we argue that $det(\bar{U}^{-1}) = \pm 1$. However, this follows by the fact that determinant is multiplicative (HW6):

$$det(\bar{U}) \cdot det(\bar{U}^{-1}) = det(\bar{I}_m) = 1$$

           where $\bar{I}_m$ denotes the m-dimensional identity matrix.
    - Given Lemma 1, let's prove Theorem 1:

- $\Leftarrow$ Suppose $\bar{B}_2 = \bar{B}_1\bar{U}$ for some unimodular matrix $\bar{U}$, which implies $\Lambda(\bar{B}_2) \subseteq \Lambda(\bar{B}_1)$. Since $\bar{B}_1 = \bar{B}_2\bar{U}^{-1}$ and since $\bar{U}^{-1}$ is an integer matrix by Lemma 1, we also have $\Lambda(\bar{B}_1) \subseteq \Lambda(\bar{B}_2)$. Together, $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$.
- $\Rightarrow$ Suppose $\bar{B}_1 := (\bar{b}_{1,1}, \cdots, \bar{b}_{1,m}), \bar{B}_2 := (\bar{b}_{2,1}, \cdots, \bar{b}_{2,m}) \in \mathbb{R}^{n\times m}$ be two bases such that $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$. This implies, in particular, that $\bar{b}_{1,1}, \cdots, \bar{b}_{1,m} \in \Lambda(\bar{B}_2)$ and vice versa. Thus, by definition of lattices, there exists two *integer* matrices $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m\times m}$ such that $\bar{B}_2 = \bar{B}_1\bar{M}_1$ and $\bar{B}_1 = \bar{B}_2\bar{M}_2$. Combining the two equations, we have

  $$\bar{B}_1 = \bar{B}_1\bar{M}_1\bar{M}_2$$

  and thus $\bar{B}_1(\bar{I}_m - \bar{M}_1\bar{M}_2) = \bar{0}_{m\times m}$, where $\bar{0}_{m\times m}$ is the $m \times m$ matrix with zero entries. Since $\bar{B}_1$ is a basis and its column vectors are linearly independent, it follows that $\bar{M}_1\bar{M}_2 = \bar{I}_m$ and thus

  $$det(\bar{M}_1\bar{M}_2) = 1.$$

  Since $\bar{M}_1$ and $\bar{M}_2$ are both integer matrices, their determinants are also integers and thus only way the above equation is satisfied (since $det(\bar{M}_1\bar{M}_2) = det(\bar{M}_1)det(\bar{M}_2)$) is when $det(\bar{M}_1) = det(\bar{M}_2) = \pm 1$.

- **HW7.** Show that if $\bar{B}_1$ and $\bar{B}_2$ are bases of the same lattice, then $det(\Lambda(\bar{B}_1)) = det(\Lambda(\bar{B}_2)$.
  - Hint: use Theorem 1.

2026s-CS758-Lecture03

Topics

- Characterising lattice bases using unimodular matrices
- Lattice determinant

  Sources
- [1]: COURSE-Micciancio07, Lecture 02 (PDF)

  Plan
- Recall previous lecture. We discussed:
  - Two definitions of lattices: a topological definition and an algebraic/geometric definition
  - Discussed differences between lattices and vectors spaces: in particular, Gram-Schmidt works for vector spaces, but not lattices
- Plan for today's lecture:
  - Characterise lattice bases using unimodular matrices
  - Learn first lattice property: the determinant
- **Characterising Lattice Bases**
  - Recall from last lecture that we saw that some lattices don't have orthogonal basis and we saw that, like vector space bases, lattice bases are not unique. Let's try to characterise lattices bases. To this end, we need to first define unimodular matrices.
  - **Definition 1.** A (square) matrix $\bar{U}$ is unimodular if it has *integer* entries and $det(\bar{U}) = \pm 1$.
    - Intuitively, a unimodular matrix preserves "volume" of the fundamental parallelepiped.
  - **Example 1.** The following are examples of unimodular matrices:
    - Permutation matrices (determinant 1), obtained by swap columns of the identity matrix:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

    - Reflection matrix (determinant -1), obtained by multiplying columns of identity matrix with -1:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

- Certain shear matrices (determinant 1):

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \pm s \\ 0 & 0 & 1 \end{pmatrix}$$

- **HW1.** Show that any unimodular matrix can be obtained by applying the following three column operations to the identity matrix:
  1. Permute: swap two columns
  2. Reflect: multiply a column by -1
  3. Shear: add an integer multiple of one column to another column
- **Theorem 1.** ([2], Theorem 2) Let $\bar{B}_1, \bar{B}_2 \in \mathbb{R}^{n \times m}$ be two bases. Then $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$ iff there exists a unimodular matrix $\bar{U} \in \mathbb{Z}^{m \times m}$ such that $\bar{B}_2 = \bar{B}_1 \bar{U}$.
  - **PQ1.** Given Theorem 1, how do you enumerate all bases for a lattice Λ?
  - **Proof of Theorem 1.**
    - We first prove the following lemma about unimodular matrices:
    - **Lemma 1.** If $\bar{U} \in \mathbb{Z}^{m \times m}$ is a unimodular matrix, then so is $\bar{U}^{-1}$
    - **Proof.** We split the proof into two parts:
      1. In the first part, we argue that $\bar{B} := \bar{U}^{-1}$ is an *integer* matrix. To see why, parse $\bar{V}$ as $(\bar{v}_1, \cdots, \bar{v}_m)$ and each $\bar{v}_i$ as $(v_{i,1}, \cdots, v_{i,m})^t$ and consider the equation

$$\bar{U}\bar{V} = \bar{I}_m,$$

where $\bar{I}_m$ denotes the m-dimensional identity matrix. Using Cramer's rule, the system of linear equations

$$\bar{U}\bar{v}_i = \bar{e}_i,$$

where $\bar{e}_i$ is the ith unit vector, has a unique solution $\bar{v}_i$ where

$$v_{i,j} := \frac{det((\bar{u}_1, \cdots, \bar{u}_{j-1}, \bar{e}_i, \bar{u}_{j+1}, \cdots, \bar{u}_n))}{det(\bar{U})}.$$

Since all the vectors in the numerator are integer, and $det(\bar{U}) = \pm 1$, we have that $\bar{v}_i$ is also an integer vector and thus $\bar{V} := \bar{U}^{-1}$ is also an integer vector.

      2. In the second part, we argue that $det(\bar{U}^{-1}) = \pm 1$. However, this follows by the fact that determinant is multiplicative (see HW2 below):

$$det(\bar{U}) \cdot det(\bar{U}^{-1}) = det(\bar{I}_m) = 1.$$

      - **HW2 (Determinant is multiplicative.)** For any $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m \times m}$, show that

$$det(\bar{M}_1\bar{M}_2) = det(\bar{M}_1)det(\bar{M}_2).$$

- Given Lemma 1, let's prove Theorem 1:
  - $\Leftarrow$ Suppose $\bar{B}_2 = \bar{B}_1\bar{U}$ for some unimodular matrix $\bar{U}$. This implies $\bar{B}_2 \subseteq \Lambda(\bar{B}_1)$ and thus $\Lambda(\bar{B}_2) \subseteq \Lambda(\bar{B}_1)$. Conversely, since $\bar{B}_1 = \bar{B}_2\bar{U}^{-1}$ and since $\bar{U}^{-1}$ is an integer matrix by Lemma 1, we also have $\bar{B}_1 \subseteq \Lambda(\bar{B}_2)$ and thus $\Lambda(\bar{B}_1) \subseteq \Lambda(\bar{B}_2)$. Taken together, $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$.
  - $\Rightarrow$ Suppose $\bar{B}_1 := (\bar{b}_{1,1}, \cdots, \bar{b}_{1,m}), \bar{B}_2 := (\bar{b}_{2,1}, \cdots, \bar{b}_{2,m}) \in \mathbb{R}^{n\times m}$ be two bases such that $\Lambda(\bar{B}_1) = \Lambda(\bar{B}_2)$. This implies, in particular, that $\bar{b}_{1,1}, \cdots, \bar{b}_{1,m} \in \Lambda(\bar{B}_2)$ and vice versa. Thus, by definition of lattices, there exists two *integer* matrices $\bar{M}_1, \bar{M}_2 \in \mathbb{Z}^{m\times m}$ such that $\bar{B}_2 = \bar{B}_1\bar{M}_1$ and $\bar{B}_1 = \bar{B}_2\bar{M}_2$. Substituting the former into the latter, we have

    $$\bar{B}_1 = \bar{B}_1\bar{M}_1\bar{M}_2$$

    and thus

    $$\bar{B}_1(\bar{I}_m - \bar{M}_1\bar{M}_2) = \bar{0}_{m\times m},$$

    where $\bar{0}_{m\times m}$ is the $m \times m$ matrix with zero entries. Since $\bar{B}_1$ is a basis and its column vectors are linearly independent, it follows that $\bar{M}_1\bar{M}_2 = \bar{I}_m$ and thus

    $$det(\bar{M}_1\bar{M}_2) = 1.$$

    Since $\bar{M}_1$ and $\bar{M}_2$ are both integer matrices, their determinants are also integers and thus only way the above equation is satisfied (by HW2) is when $det(\bar{M}_1) = det(\bar{M}_2) = \pm 1$.
- **HW3 (Determinant is basis-independent).** Show that if $\bar{B}_1$ and $\bar{B}_2$ are bases of the same lattice, then $det(\Lambda(\bar{B}_1)) = det(\Lambda(\bar{B}_2)$.
  - Hint: use Theorem 1.
- **HW4 (Existence vs search).** Theorem 1 proves the *existence* of a unimodular matrix $\bar{U}$. Can you find the $\bar{U}$ efficiently?

- **Lattice Determinant.** In this section we discuss our first fundamental lattice property: the *determinant*. Geometrically, it captures *density* of lattice points in its ambient space: if the determinant is small then the lattice is dense and vice versa. Before defining the determinant, we define the fundamental parallelepiped (or region) of a lattice
  - **Definition 2.** (Fundamental parallelepiped) Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n\times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

    $$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i\bar{b}_i : 0 \le x_i < 1 \right\}$$

  - Geometrically, when $m = n$, the whole space $\mathbb{R}^n$ can be obtained by "tiling" using $P(\bar{B})$.

- **Definition 3.** (Determinant) Let $\bar{B} \in \mathbb{R}^{n \times m}$ be a basis. The determinant of $\Lambda(\bar{B})$ is the m-dimensional volume of $P(\bar{B})$:

$$det(\Lambda(\bar{B})) = vol(P(\bar{B})).$$

  - Note that when $m = n$, the definition is the same as that of matrix determinant.
- The following homework asks you to prove a loose upper bound on the determinant. Then you will show that the determinant can be exactly computed using the Gram-Schmidt algorithm, which we discussed Lecture 02, and discussed to be efficient.
  - **HW4 (Hadamard inequality).** Show that $det(\Lambda(\bar{B})) \leq \Pi_{i=1}^{m} ||\bar{b}_i||$.
  - **HW5 (Determinant is efficiently computable).** Let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*) \in \mathbb{R}^{n \times m}$ be the basis obtained by running the Gram-Schmidt algorithm on $\bar{B}$ in Definition 5. Then

$$det(\Lambda(\bar{B})) = \Pi_{i=1}^{m} ||\bar{b}_i^*||.$$

- We'll now see how to reduce computing the lattice determinant to matrix determinant, which is much more efficient (and stable) than Gram-Schmidt.
- **Theorem 2 (Theorem 6, in [2]).** For any lattice basis $\bar{B} \in \mathbb{R}^{n \times m}$,

$$det(\Lambda(\bar{B}) = \sqrt{det(\bar{B}^t \bar{B})}.$$

- *Proof.* First observe that the Gram-Schmidt algorithm is captured by the matrix relation

$$\bar{B} = \bar{B}^* \bar{T},$$

  where $\bar{T} \in \mathbb{R}^{m \times m}$ is the upper triangular matrix with 1s on the diagonal and for all $1 < j < i \leq m$, the value

$$\mu_{i,j} := \frac{\langle \bar{b}_2, \bar{b}_1^* \rangle}{\langle \bar{b}_1^*, \bar{b}_1^* \rangle}$$

  in cell (j, i). For example, for $m = 4$,

$$\bar{T} := \begin{pmatrix} 1 & \mu_{2,1} & \mu_{3,1} & \mu_{4,1} \\ 0 & 1 & \mu_{3,2} & \mu_{4,2} \\ 0 & 0 & 1 & \mu_{4,3} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

If follows that

$$\begin{aligned}\sqrt{det(\bar{B}^t\bar{B})} &= \sqrt{det((\bar{B}^*\bar{T})^t\bar{B}^*\bar{T})} \\ &= \sqrt{det(\bar{T}^t{\bar{B}^*}^t\bar{B}^*\bar{T})} \\ &= \sqrt{det({\bar{B}^*}^t\bar{B}^*)} \\ &= \sqrt{\Pi_i\langle\bar{b}_i^*,\bar{b}_i^*\rangle} \\ &= \sqrt{\|\bar{b}_i^*\|^2} \\ &= det(\Lambda(\bar{B}))\end{aligned}$$

- In the following homework, you'll show that every integer lattice contains an orthogonal sub-lattice:
- **HW6.** For any non-singular $\bar{B} \in \mathbb{Z}^{m\times m}$, $det(\bar{B})\mathbb{Z}^n \subseteq \Lambda(\bar{B})$.
    - Hint: use Cramer's rule.

2026s-CS758-Lecture04

Topics

- Minimum distance
- Blichtfeldt's and Minkowski's theorems

Sources

1. CS206A, Lecture 03 (COURSE-Micciancio07)
2. ToTD article
3. Paper 1: PPP-Completeness with Connections to Cryptography (FOCS-SotZamZir18 STUB)
4. Paper 2: Reducitons in PPP (IPL-BJPPR19 STUB)

Plan

- In the previous lecture
  - We characterised lattice bases using *unimodular matrices*
  - We also defined the first fundamental parameter associated with a lattice: its *determinant*
- In today's lecture
  - We will encounter another fundamental parameter in the form of *minimum distance*
    - Interesting from coding theory and cryptography point of view
  - We will estimate the minimum distance from below and above, and in this process state and prove two interesting theorems about lattices: Blichtfeldt's theorem and Minkowski's theorem
  - Finally, we will use Minkowski's theorem to prove Fermat's theorem on sum of two squares
- Minimum distance
  - Two definitions, equivalent because lattice is a subgroup of $(\mathbb{R}, +)$
    - **Definition 1.** The minimum distance of a lattice Λ is defined as

$$\lambda_1(\Lambda) := \inf_{\bar{y}_1, \bar{y}_2 \in \Lambda, \bar{y}_1 \neq \bar{y}_2} \|\bar{y}_1 - \bar{y}_2\|$$

    - **Definition 2.** The minimum distance of a lattice Λ is defined as

$$\lambda_1(\Lambda) := \inf_{0^n \neq \bar{y} \in \Lambda} \|\bar{y}\|$$

    - **Example 1 (from [2]).** For the lattice Λ generated by $(2, 0)^t$ and $(1, 1)^t$, $\lambda_1(\Lambda) = \sqrt{2}$.
    - **HW1.** Show that Definition 1 is equivalent to Definition 2.
  - Motivation:

- Cryptography: cryptography can be based on the hardness of *estimating* minimum distance (GapSVP) *in the worst case*!
- Coding theory: (random) lattices are good (linear) error-correcting codes:
  - Lattice basis $\bar{B} \in \mathbb{Z}^{n \times m}$ (recall that $m < n$) corresponds to the generator matrix, and thus the message space consists vectors in $\mathbb{Z}^m$ and the lattice $\Lambda(\bar{B})$ generated by $\bar{B}$ are the codewords.
  - Minimum distance corresponds to the minimum Hamming distance of the code, which corresponds to the unique decoding radius
    - That is, if the corrupted codeword $\bar{y}'$ is in a ball of radius λ_1 of some some codeword (i.e, a lattice point), then unique decoding is possible
  - Therefore, it is desirable to have a *large* minimum distance

- Lower Bound
  - **Claim 1 ([1], Proposition 2).** For a lattice $\Lambda(\bar{B})$, let $\bar{B}^* := (\bar{b}_1^*, \cdots, \bar{b}_m^*)$ denote the Gram-Schmidt orthogonalisation of $\bar{B}$. Then

$$\lambda_1(\Lambda(\bar{B})) \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|$$

  - *Proof.*
    - Let $\bar{B}\bar{x}$, $\bar{x} \in \mathbb{Z}^m \setminus \{0^n\}$, be any non-zero lattice vector and let $1 \leq I \leq m$ be the largest index such that $x_I \neq 0$ (i.e., $\bar{x}$ is of the form $(x_1, \cdots, x_I, 0, \cdots, 0)^t$). We first show that

$$\|\bar{B}\bar{x}\| \geq \|\bar{b}_I^*\| \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|.$$

      To see why observe that

$$\begin{aligned} \langle \bar{B}\bar{x}, \bar{b}_I^* \rangle &= \sum_{1 \leq i \leq I} \langle x_i \bar{b}_i, \bar{b}_I^* \rangle \\ &= x_I \langle \bar{b}_I, \bar{b}_I^* \rangle \\ &= x_I \|\bar{b}_I^*\|^2, \end{aligned}$$

      where the second and third equalities follow by orthogonality of $\bar{b}_I^*$ and $\bar{b}_i$s for $i < I$. Now, by Cauchy-Schwartz inequality and since $|x_I| \geq 1$, we get

$$\|\bar{B}\bar{x}\| \cdot \|\bar{b}_I^*\| \geq |\langle \bar{B}\bar{x}, \bar{b}_k^* \rangle| = |x_I| \cdot \|\bar{b}_I^*\|^2 \geq \|b_I^*\|^2$$

      and thus

$$\|\bar{B}\bar{x}\| \geq \|\bar{b}_I^*\| \geq \min_{1 \leq i \leq m} \|\bar{b}_i^*\|.$$

  - Using Claim 1, it can be shown that there exists a lattice point $\bar{y}$ such that $\|\bar{y}\| = \lambda_1$:
    - **HW2.** Show that the infimum in Definitions 1 and 2 can be replaced by minimum.
- Upper Bound

- To upper bound λ_1, we prove the existence of a lattice point of certain length determined by its determinant (and dimension) and hence independent of the basis
- **Theorem 1 ([1], Corollary 6).** For any full-rank lattice $\Lambda(\bar{B})$, there exists a non-zero lattice point $\bar{y}$ such that

$$\|\bar{y}\| \leq \sqrt{n} \cdot det(\bar{B})^{1/n}$$

- Interestingly, Theorem 1 boils down to the pigeonhole principle! We will prove the claim in three steps:
  1. First, we will prove Blichtfeldt's theorem, which informally says that any large-enough region $\mathcal{S}$ of the ambient space must have two (not-necessarilty-lattice) points such that their difference is a lattice point:
     - **Theorem 2 ([1], Theorem 3).** For a full-rank lattice $\Lambda(\bar{B})$ and a set $\mathcal{S} \subseteq \mathbb{R}^n$, if

       $$vol(\mathcal{S}) > vol(P(\bar{B})) =: det(\bar{B}),$$

       then $\mathcal{S}$ contains two points $\bar{z}_1 \neq \bar{z}_2$ such that $\bar{z}_1 - \bar{z}_2 \in \Lambda(\bar{B})$.
     - **Example 2.** Consider Λ from Example 1, whose fundamental parallelepiped has volume 2 (units)
     - Proof
       1. Let's *partition* $\mathcal{S}$ using *translated copies* of the fundamental paralellepiped $P(\bar{B})$. That is,

          $$\mathcal{S} = \cup_{\bar{y} \in \Lambda(\bar{B})} \mathcal{S}_{\bar{y}},$$

          where for $\bar{y} \in \Lambda(\bar{B})$ we define

          $$\mathcal{S}_{\bar{y}} := \mathcal{S} \cap (\bar{y} + P(\bar{B})).$$

          Note that

          $$vol(\mathcal{S}) = \sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}}).$$

       2. Next, consider these partitions *translated* to the fundamental parallelepiped, i.e., the sets

          $$\mathcal{S}_{\bar{y}} - \bar{y} = (\mathcal{S} - \bar{y}) \cap P(\bar{B}) \subseteq P(\bar{B}).$$

       3. By pigeonhole principle the translated partitions $\mathcal{S}_{\bar{y}} - \bar{y}$ cannot be mutually disjoint since

          $$\sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}} - \bar{y}) = \sum_{\bar{y} \in \Lambda(\bar{B})} vol(\mathcal{S}_{\bar{y}}) = vol(\mathcal{S}) > vol(P(\bar{B})).$$

Therefore there exist two vectors $\bar{y}_1 \neq \bar{y}_2 \in \Lambda(\bar{B})$ such that

$$(\mathcal{S}_{\bar{y}_1} - \bar{y}_1) \cap (\mathcal{S}_{\bar{y}_2} - \bar{y}_2) \neq \emptyset.$$

4. Let's call such an (non-lattice) "collision" vector $\bar{z}$ and define

$$\bar{z}_1 := \bar{z} + \bar{y}_1 \in \mathcal{S}_{\bar{y}_1}, \bar{z}_2 := \bar{z} + \bar{y}_2 \in \mathcal{S}_{\bar{y}_2}$$

to be their "pre-image" in the original partitions. Now note that

$$\bar{z}_1 - \bar{z}_2 = \bar{y}_1 - \bar{y}_2 \in \Lambda(\bar{B}).$$

2. Second, we will use Blichtfeldt's theorem to prove Minkowski convex-body theorem
   - **Theorem 3 ([1], Corollary 4).** Consider a lattice $\Lambda(\bar{B})$. If $\mathcal{C} \subseteq \mathbb{R}^n$ is a *convex*, *symmetric* set such that

     $$vol(\mathcal{C}) > 2^n det(\bar{B}),$$

     then $\mathcal{C}$ contains a (non-zero) *lattice point*. Recall that:
     - If $\mathcal{C}$ is *convex* then for any $\bar{z}_1, \bar{z}_2 \in \mathcal{C}$, $(\bar{z}_1 + \bar{z}_2)/2 \in \mathcal{C}$.
     - $\mathcal{C}$ is symmetric if $\bar{z} \in \mathcal{C} \Rightarrow -\bar{z} \in \mathcal{C}$
   - *Proof.* Consider the set

     $$\mathcal{C}/2 := \{\bar{z} : 2\bar{z} \in \mathcal{C}\},$$

     and note that

     $$vol(\mathcal{C}/2) = vol(\mathcal{C})/2^n > det(\bar{B}).$$

     Applying Theorem 1 with $\mathcal{S} := \mathcal{C}/2$, there exist $\bar{z}_1 \neq \bar{z}_2 \in \mathcal{C}/2$ such that

     $$\bar{y} := \bar{z}_1 - \bar{z}_2 \in \Lambda(\bar{B}) \setminus \{0^n\}.$$

     We now claim that $\bar{y}$ is also in $\mathcal{C}$ by symmetry and convexity:
     1. By definition of $\mathcal{C}/2$, $2\bar{z}_1, 2\bar{z}_2 \in \mathcal{C}$.
     2. By symmetry $-2\bar{z}_2 \in \mathcal{C}$.
     3. By convexity

     $$\mathcal{C} \ni (2\bar{z}_1 - 2\bar{z}_2)/2 = \bar{z}_1 - \bar{z}_2 = \bar{y}.$$

3. Finally, we will use Minkowski's theorem to prove Theorem 1.
   - **HW3.** Using Theorem 3, show that for any full-rank lattice $\Lambda(\bar{B})$, there exists non-zero lattice point $\bar{y}$ such that

     $$\|\bar{y}\|_\infty \leq det(\bar{B})^{1/n},$$

     where $\|\cdot\|_\infty$ denotes the ℓ_∞ norm, i.e.,

     $$\ell_\infty(\bar{y}) := \max_{1 \leq i \leq n} |y_i|.$$

        - **HW4.** Show that $\|\bar{y}\| \leq \sqrt{n} \cdot \|\bar{y}\|_\infty$.
    - Estimates on λ_1 via Minkowski's theorem need not be tight
        - **Example 1.** Consider the 2D lattice generated by $(1,0)^t$ and $(0,N)^t$ for a large $N \in \mathbb{N}$. The lattice contains a short vector $(1,0)^t$ of length 1, but its determinant is N and Minkowski's bound $\sqrt{2N}$ is much larger than 1.
    - Conversely, there are lattices where Minkowski's bound is close to tight
        - **HW4.** For any n, there is an n dimensional lattice Λ_n such that

$$\lambda_1(\Lambda_n) \geq c\sqrt{n} \cdot det(\Lambda_n)^{1/n}.$$

    - Computational problems associated with Theorems 2 and 3:
        - Theorems 2 and 3 are existence theorems, and have natural search counterparts [3,4]:
            - MINKOWSKI
                - Inputs: full-rank lattice $\Lambda(\bar{B})$ of dimension n of determinant at least $\sqrt{n} \cdot 2^n$
                - Solution: non-zero lattice point in $[-1,1]^n$
            - BLICHTFELDT
                - Inputs: full-rank lattice $\Lambda(\bar{B})$ of dimension n and $\mathcal{S} \subseteq \mathbb{R}^n$ such that $vol(\mathcal{S}) > vol(P(\bar{B}))$ described using a succinct a circuit
                - Solution: $\bar{y}_1 \neq \bar{y}_2 \in \mathcal{S}$ such that $\bar{y}_1 - \bar{y}_2 \in \Lambda(\bar{B})$
        - Known to be in a complexity class called $\mathbf{PPP}$, which contain problems "reducible to " pigeonhole principle
    - **Open Question 1.** Are MINKOWSKI and BLICHTFELDT *complete* for $\mathbf{PPP}$?
- Application
    - **Theorem 3 ([1], Theorem 7).** For every prime $p \equiv 1 \bmod 4$, there exists integers $a, b \in \mathbb{Z}$ such that $p = a^2 + b^2$.
    - *Proof.* See [1]

2026s-CS758-Lecture05

Topics

- Successive minima
- Shortest-vector problem (SVP) and its friends
- Lagrange's algorithm for 2D-SVP

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 2
2. Intro to Lattice Algorithms and Cryptography, Lecture 3
3. Intro to Lattice Algorithms and Cryptography, Lecture 4
4. The LLL Algorithm: Survey and Algorithms, Chapter 02

Plan

- Plan/Recall
  - In the last lecture,
    - We defined the minimum distance $\lambda_1(\Lambda)$ of a lattice Λ and saw how it is relevant to cryptography and coding theory
    - Then we went on to establish upper and lower bounds for the minimum distance. In particular, we saw Minkowski's first theorem, which upper bounds λ_1 in terms of the determinant (which is basis-independent):

$$\lambda_1(\Lambda(\bar{B})) \leq \sqrt{n} \cdot det(\bar{B})^{1/n}$$

      - Essentially a consequence of pigeonhole principle via Blichtfeldt's theorem and Minkowski's convex-body theorem
  - In today's lecture:
    - We first define *successive minima*, which generalises the notion of minimum distance
    - Then we will introduce computational problems related to minimum distance: the shortest vector problem (SVP) and its approximate variant
    - In general SVP is a hard problem, we'll see that for the 2D case, there is an efficient algorithm: the Lagrange's algorithm. Along the way, we will learn reduced bases
- Successive Minima
  - Recall the definition of λ_1: it can be seen as the radius of the smallest ball that contains a non-zero lattice point
  - Informally, λ_i is the radius of the smallest ball that contains i *linearly-independent* lattice points

- **Example 1.** Consider $\Lambda := \Lambda((1,0)^t, (3,4)^T)$. Note that $\lambda_1(\Lambda) = 1$ since $(1,0)^t \in \Lambda$. Note that $(2,0)^t$ is also in Λ, but it is linearly dependent on $(1,0)^t$. It seems that $\lambda_2(\Lambda) = 4$ since $(0,4) \in \Lambda$.
- More formally:
  - **Definition 1.** For $1 \leq i \leq m$, we define

    $$\lambda_i(\Lambda) := \inf_{r \geq 0} dim(\Lambda \cap B_r^n) \geq i,$$

    where B_r^n is the n-dimensional ball of radius r
    - Note that $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_m$
  - **HW1.**
    - Let $\Lambda \subset \mathbb{R}^n$ be a m-rank lattice. Show that there exists vectors $\bar{y}_1, \cdots, \bar{y}_m \in \Lambda$ such that $\|\bar{y}_i\| = \lambda_i$.
    - Does Λ have a basis $\bar{b}_1 \cdots, \bar{b}_m$ such that $\|\bar{b}_i\| = \lambda_i$? This is true for Λ in Example 1, but is it true in general?
- Minkowski also proved the following theorem bounding the *geometric mean* of successive minima:
  - **Theorem 1 (Minkowski's second theorem, see [1], Theorem 21).** For any full-rank lattice Λ of dimension n,

    $$(\Pi_{i=1}^n \lambda_i(\Lambda))^{1/n} \leq \sqrt{n} \cdot det(\Lambda)^{1/n}$$

    - Note that Theorem 2 is a stronger bound than Minkowski's first theorem since the mean is at least λ_1.

- Shortest-Vector Problem and its Friends
  - We are now ready to define some computational problems on lattices
    - **Definition 2 (SVP: Shortest Vector Problem, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ
      - *Solution.* A vector $\bar{y} \in \Lambda$ such that $\|\bar{y}\| = \lambda_1(\Lambda)$
    - **Definition 3 (SVP$_\alpha$: Approximate SVP, [3], Definition 1).**
      - *Input.* A basis $\bar{B}$ of a lattice Λ of dimension n and an approximation factor $\alpha(\cdot)$
      - *Solution.* A vector $y \in \Lambda$ such that $\|\bar{y}\| \leq \alpha(n)\lambda_1(\Lambda)$
    - **Definition 4 (GapSVP$_\gamma$: gap SVP, [4], Definition 1).**
      - *Input.* A basis $\bar{B}$ of lattice Λ, a distance $d \in \mathbb{Q}$ and an gap factor $\gamma(n)$
      - *Decision.* Is $\lambda_1(\Lambda) > \gamma(n)d$ or $\lambda_1(\Lambda) \leq d$?
    - **Definition 5 (SIVP$_\alpha$: Approximate Shortest Independent Vector Problem, [4], Definition 2).**
      - *Input.* A basis $\bar{B}$ of lattice Λ and an approximation factor $\alpha(\cdot)$
      - *Solution.* m linearly-independent lattice vectors $\bar{y}_1, \cdots, \bar{y}_m$ such that $\|\bar{y}_i\| \leq \alpha(n)\lambda_n(\Lambda)$
  - What do we know about these problems? (Source)

- SVP
  - Ajtai showed that SVP is NP-hard using a decision-to-search (randomised) reduction from *any* NP problem to SVP
    - Priyanshu will cover basics of complexity theory in Recitation 2.
  - We don't know how to base cryptography on SVP
- SVP$_\alpha$
  - Khot showed that SVP is NP-hard for any constant approximation factor, i.e, $\alpha := O(1)$
  - For approximation factors $\alpha(n) := \sqrt{n/\log(n)}$ and $\alpha(n) := O(\sqrt{n})$, SVP$_\alpha$ was shown to be in $\mathbf{coNP}$ and $\mathbf{coAM}$ respectively by Aharonov and Regev and Goldreich and Goldwasser
  - For approximation factors $\alpha(n) := n$, it seems we can do cryptography (I was wrong!).
  - In the next lecture, we will see the LLL algorithm, which solves SVP$_\alpha$ for $\alpha := 2^{O(n)}$.
- Here is a very nice schematic by Vinod Vaikuntanathan capturing all of the above:

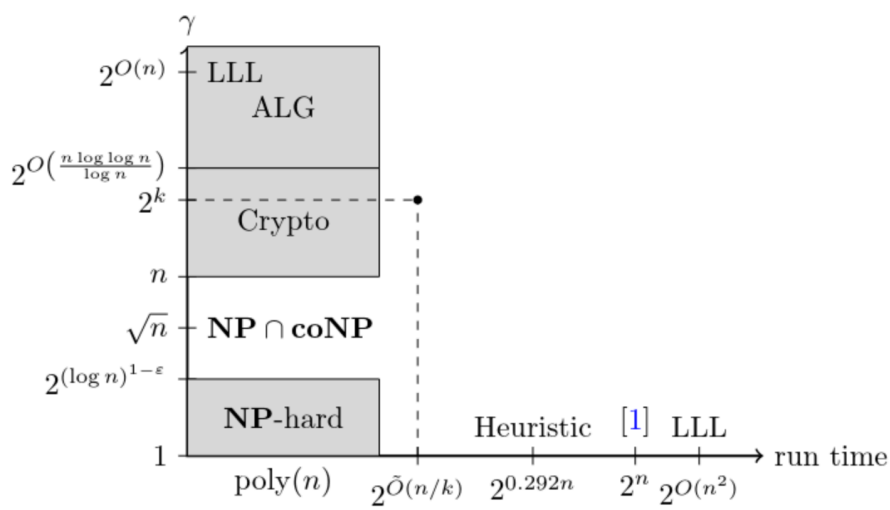

Figure 1: The complexity landscape of SVP$_\gamma$.

- GapSVP$_\gamma$
  - We can build cryptography from GapSVP$_\gamma$ for $\gamma(n) := O(n)$: see the work of Micciancio and Regev
- SIVP$_\alpha$
  - Ajtai showed how to base cryptography on approximate SIVP for $\alpha := poly(n)$. We will see this result in Module II

- Lagrange's algorithm for 2D-SVP
  - In this section we will see that although (exact) SVP is $\mathbf{NP}$-hard, the hardness only really kicks in in higher dimensions. That is, we will see a polynomial-time algorithm for SVP when $n = 2$, called Lagrange's algorithm. In this process we will

introduce the notion of a "reduced" basis. The techniques used will be useful in the next lecture when we discuss the celebrated LLL algorithm for *approximate* SVP.

- *Reduced basis.* Informally a basis is "reduced" if consists of short, almost orthogonal vectors. There are several notions of reduced. In this lecture, we will consider "Lagrange" reduced basis for $\mathbb{R}^2$, while in the next lecture we will consider "LLL-reduced" basis for $\mathbb{R}^n$.
  - **Definition (Wristwatch Condition).** A basis $(\bar{b}_1, \bar{b}_2)$ for a lattice $\Lambda \subset \mathbb{R}^2$ is Lagrange-reduced iff

    $$\|\bar{b}_1\| \leq \|\bar{b}_2\|$$

    and

    $$|\langle \bar{b}_1, \bar{b}_2 \rangle| \leq \|\bar{b}_1\|^2/2.$$

    - Geometrically, it looks like a wrist-watch: see figure.
  - In the following claim, we show that a Lagrange-reduced basis is in fact the "shortest basis"!
  - **Claim 1.** For a Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2)$, $\|\bar{b}_1\| = \lambda_1(\Lambda)$ and $\|\bar{b}_2\| = \lambda_2(\Lambda)$.
  - *Proof.*
    1. First we show that $\|\bar{y}\| \geq \|\bar{b}_1\|$ for any non-zero $\bar{y} \in \Lambda$. Since $\bar{y} \in \Lambda$, there exists $x_1, x_2 \in \mathbb{Z}$ such that $\bar{y} = x_1\bar{b}_1 + x_2\bar{b}_2$. Now we have,

       $$\begin{aligned} \|\bar{y}\|^2 &= \|x_1\bar{b}_1 + x_2\bar{b}_2\|^2 \\ &= x_1^2\|\bar{b}_1\|^2 + 2x_1x_2\langle \bar{b}_1, \bar{b}_2 \rangle + x_2^2\|\bar{b}_2^2\| \\ &\geq x_1^2\|\bar{b}_1^2\| - x_1x_2\|\bar{b}_1\|^2 + \|\bar{b}_2^2\| \\ &\geq (x_1^2 - x_1x_2 + x_2^2)\|\bar{b}_1\|^2 \\ &\geq \|\bar{b}_1\|^2, \end{aligned}$$

       where
       - The first inequality holds by the second clause of wristwatch condition, i.e., $\langle \bar{b}_1, \bar{b}_2 \rangle \geq -\|\bar{b}_1\|^2/2$,
       - The second inequality holds by its first clause, and
       - The last inequality holds since $\bar{y}$ is non-zero and both $x_1 = 0$ and $x_2 = 0$ cannot hold
    2. Next, we show that $\|\bar{y}\| \geq \|\bar{b}_2\|$ whenever $x_2 \neq 0$ (i.e., $\bar{y}$ is not linearly dependent on $\bar{b}_1$.
       - **HW2.** Prove the second part of Claim 1.
- *The algorithm.*
  - The algorithm is similar to Euclid's GCD algorithm, which we now recall:
    - **Algorithm 1**:
      - *Input*: $a, b \in \mathbb{Z}$
      - *Output*: $gcd(a, b)$
      - Algorithm:

1. If $|a| \leq |b|$ then swap a and b
2. Else, while $b \neq 0$ do:
    1. Compute $r := a - \lfloor a/b \rceil b$, where $\lfloor \cdot \rceil$ is the function that rounds to the nearest integer (say breaking ties upwards)
    2. Set $a := b$ and $b := r$
3. Output a

- **Example 2.** Running Algorithm 1 on $(56, 315)$, results in the sequence $(315, 56)$, $(56, 35)$, $(35, 21)$, $(21, 14)$, $(14, 7)$, $(7, 0)$

- The idea is to carry out Gram-Schmidt *with rounding* and swap the resulting basis vectors as long as $\bar{b}_1$'s length is greater than $\bar{b}_2$
    - **Algorithm 2**:
        - *Input*: Basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - *Output*: Lagrange-reduced basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - Algorithm:
            1. If $\|\bar{b}_1\| < \|\bar{b}_2\|$ then swap $\bar{b}_1$ and $\bar{b}_2$
            2. Else, repeat
                1. Compute $\bar{r} := \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2$, where if you recall $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \langle \bar{b}_2, \bar{b}_2 \rangle$
                2. Set $\bar{b}_1 := \bar{b}_2$ and $\bar{b}_2 := \bar{r}$.
                Until $\|\bar{b}_1\| \leq \|\bar{b}_2\|$.
            3. Output $(\bar{b}_1, \bar{b}_2)$
    - **Example 3.** On running Algorithm 2 on $((1,5)^t, (6,21)^t)$ results in the sequence $((6,21)^t, (1,5)^t)$, $((1,5)^t, (2,1)^t)$ and $(2,1)^t, (-1,4)^t)$. See figure
- **Theorem 2.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations and outputs a Lagrange-reduced basis.
- *Proof.*
    - **Claim 2.** Algorithm 2 outputs a Lagrange-reduced basis.
    - *Proof.*
        - **Pop-Quiz 1.** Why does the output forms a basis of Λ? Because we are only doing the column operations corresponding to unimodular matrix
        - **Pop-Quiz 2.** Why does the wristwatch condition hold?
            - First, note that Algorithm 1 *terminates* since $\|\bar{b}_1\|$ diminishes in each iteration: we are replacing $\bar{b}_1$ with the shorter of the basis vectors, i.e., $\bar{b}_2$, and $\bar{b}_2$ is shrinking since we are replacing it with "rounded" orthogonal projections of $\bar{b}_1$ (similar to Gram-Schmidt). Since termination can only happen when $\|\bar{b}_1\| \leq \|\bar{b}_2\|$, the first wristwatch condition holds.

- To see why, let's denote the basis at the end of each iteration by $(\bar{b}_1', \bar{b}_2')$ observe by construction that

$$\begin{aligned}\langle \bar{b}_1', \bar{b}_2' \rangle &= \langle \bar{b}_2, \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2 \rangle \\ &= \langle \bar{b}_1, \bar{b}_2 \rangle - \lfloor \mu_{1,2} \rceil \|\bar{b}_2\|^2\end{aligned}$$

  Showing that the above expression is at most $\|\bar{b}_1'\|^2/2 = \|\bar{b}_2\|^2/2$ is the same as showing

$$\langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2 - \lfloor \mu_{1,2} \rceil \leq 1/2.$$

  However, this is true since $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_2\|^2$ and definition of rounding.

- **Claim 3.** Algorithm 2 runs for $O(\log(\|\bar{b}_1\| / \sqrt{det(\Lambda)})$ iterations.
  - **HW3.** Show that the length of $\bar{b}_1$ reduces by a constant factor in each iteration. What exactly is the constant

2026s-CS758-Lecture06

Topics

- Hermite's bound and reduction algorithm
- Weak LLL algorithm

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 3
2. Intro to Lattice Algorithms and Cryptography, Lecture 4
3. The LLL Algorithm: Survey and Applications, Chapter 02 (PDF)

Plan

- Recall/Plan
  - In the last lecture:
    - We saw Lagrange's reduction, a poly-time *exact* algorithm for SVP in 2D. Recall the algorithm:
      - **Algorithm 1**:
        - *Input*: Basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$
        - *Output*: A basis $(\bar{b}_1, \bar{b}_2) \in \mathbb{R}^{2\times 2}$ with $\|\bar{b}_1\| \leq \|\bar{b}_2\|$ and Gram-Schmidt coefficient $\mu_{1,2} \leq 1/2$ (recall wristwatch figure), which implies $\|\bar{b}_1\| = \lambda_1$
        - Algorithm:
          1. If $\|\bar{b}_1\| < \|\bar{b}_2\|$ then swap $\bar{b}_1$ and $\bar{b}_2$
          2. Else, repeat
             1. Compute $\bar{r} := \bar{b}_1 - \lfloor \mu_{1,2} \rceil \bar{b}_2$, where if you recall $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \langle \bar{b}_2, \bar{b}_2 \rangle$
             2. Set $\bar{b}_1 := \bar{b}_2$ and $\bar{b}_2 := \bar{r}$.
             Until $\|\bar{b}_1\| \leq \|\bar{b}_2\|$.
          3. Output $(\bar{b}_1, \bar{b}_2)$
    - We argued that Algorithm 1 runs in polynomial time since the length of $\bar{b}_1$ reduces by at least half in each iteration.
    - A natural question to ask is what happens when you run Lagrange's reduction in higher dimensions.
  - In today's lecture:
    - We will see Hermite's reduction, a natural generalisation of Lagrange's reduction to higher dimensions that solves *approximate SVP* . The approximation factor turns out to be exponential in the dimension n. To this end we will define Hermite's constant.

    - However, we don't know whether Hermite's reduction runs in polynomial time. We partially resolve this by tweaking Hermite's reduction to obtain a *weak LLL* reduction, in which the number of *iterations* is polynomial in the dimension.
    - In the next lecture we will see how to further tweak Hermite's reduction to obtain the celebrated LLL reduction, which *can* be shown to be poly-*time*.
- Hermite's Bound
    - Recall Minkowski's first theorem, which bounds the minimum distance of a rank-n lattice Λ as

$$\lambda_1(\Lambda) \leq \sqrt{n} \cdot \det(\Lambda)^{1/n}.$$

    - An older, weaker bound on minimum distance, due to Hermite, is stated below in Theorem 1.
    - **Definition 1 (Definitions 7 and 8 in [1]).** For a full-rank lattice $\Lambda \subset \mathbb{R}^n$, we define

$$\gamma_n(\Lambda) := (\lambda_1(\Lambda)/\det(\Lambda)^{1/n})^2.$$

      Hermite's constant is defined as

$$\gamma_n := \sup_{\Lambda \text{ of rank } n} \gamma_n(\Lambda).$$

    - **Theorem 1 (Lemma 9 in [1] and Theorem 5 in [2]).** We have

$$\gamma_2 = \sqrt{4/3}$$

      and

$$\gamma_n \leq \gamma_2^{n-1}.$$

    - A couple of remarks on Theorem 1 are in order:
      1. The supremum for γ_2 is achieved by the hexagonal lattice, generated by $((0,1),(\sqrt{3/4},1/2))$: see Figure 1.
      2. A corollary to Theorem 1 is that for any rank-n lattice Λ

$$\lambda_1(\Lambda) \leq (4/3)^{(n-1)/4} \cdot \det(\Lambda)^{2/n},$$

         which is quantitatively weaker than Minkowski's bound. Nevertheless, the bound is interesting since Hermite's reduction and weak LLL reduction can be regarded as an algorithmisation of its proof, and thus yield an $2^{O(n)}$-*approximate* solution to SVP.
- Hermite's Reduction
    - Before extending Lagrange's algorithm to higher dimensions, we need to define two notions: orthogonal complement (which we already saw in Gram-Schmidt) and lattice projections.

- **Definition 2.** For a vector space V, the *orthogonal complement* of a linear subspace $W \subseteq V$ is defined as

$$W^{\perp} := \{\bar{b} \in V : \langle \bar{v}, \bar{w} \rangle \text{ for all } \bar{w} \in W\}.$$

  If W has a basis $(\bar{w}_1, \cdots, \bar{w}_k)$ then we also write $W^{\perp}$ as $(\bar{w}_1, \cdots, \bar{w}_k)^{\perp}$.
- **Notation 1.** For a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ and $1 \leq i \leq j \leq n$, we use $\pi_i(\bar{b}_j)$ to denote the *projection* of $\bar{b}_j$ on to the orthogonal complement $(\bar{b}_1, \cdots, \bar{b}_i)^{\perp}$.
  - Note that using this notation, the Gram-Schmidt basis $\bar{B}^*$ of $\bar{B}$ can be denoted by

$$(\pi_1(\bar{b}_1), \pi_2(\bar{b}_2), \cdots, \pi_n(\bar{b}_n))$$

- **Definition 3.** For a lattice Λ generated by $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ and $1 \leq i \leq j \leq n$, we define

$$\bar{B}_{i:j} : (\pi_i(\bar{b}_i), \cdots, \pi_i(\bar{b}_j))$$

  and the projected lattice $\Lambda_{i:j}$ as the lattice generated by $\bar{B}_{i:j}$, i.e.,

$$\Lambda_{i:j} := \Lambda((\pi_i(\bar{b}_i), \cdots, \pi_i(\bar{b}_j))$$

- **Example 1.** For a lattice $\Lambda((1,0,0)^t, (0,1,5)^t, (0,6,21)^t))$ (see Figure 2), $\bar{B}_{2:3}$ is $((0,1,5)^t, (0,6,21)^t)$ since $(0,1,5)^t$ and $(0,6,21)^t$ are already orthogonal to $(1,0,0)$. Hence $\Lambda_{2:3} = \Lambda(((0,1,5)^t, (0,6,21)^t))$.
- **Homework 1.** Prove that

$$\det(\Lambda_{i:j}) = \Pi_{k=i}^{j} \|\bar{b}_k^*\|.$$

- *Hermite's reduction.* We are now ready to state the algorithm, which "locally" carries out Lagrange reduction between pairs of bases in consecutive dimensions till all bases are Lagrange reduced.
- **Algorithm 2**:
  - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$
  - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq \gamma_2^{n-1} \cdot \det(\Lambda)^{2/n}$
  - Algorithm:
    1. While $\exists i \in [n]$ such that $\bar{B}_{i:i+1} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))$ is not Lagrange-reduced, do:
       1. Find unimodular $\bar{U} \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}'_i, \bar{b}'_{i+1}) := \bar{B}_{i:i+1}\bar{U} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))\bar{U}$ is Lagrange reduced
       2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}'_i, \bar{b}'_{i+1}, \bar{b}_{i+2}, \cdots, \bar{b}_n)$
    2. Output $\bar{B}$.
- **Example 2.** Consider $\Lambda := \Lambda((1,0,0)^t, (0,1,5)^t, (0,6,21)^t)$ (see Figure 3).
  - In the first iteration, $i = 2$ is a candidate since

$$\bar{B}_{2:3} := (\pi_2((0,1,5)^t), \pi_2((0,6,21)^t)) = ((0,1,5)^t, (0,6,21)^t)$$

is not Lagrange-reduced. Carrying out Lagrange reduction on $\bar{B}_{2:3}$ yields (as we saw in last lecture)

$$((0,2,1)^t,(0,-1,4)^t).$$

Thus the update step yields

$$\bar{B}=((1,0,0)^t,(0,2,1)^t,(0,-1,4)^t)).$$

- Since $(1,0,0)^t$ and $(0,2,1)^t$ are orthogonal, and $\|(1,0,0)^t\| \leq \|(0,2,1)^t\|$, they are *already* Lagrange-reduced and we're done.

- **Theorem 2.** Algorithm 2 *terminates* and *outputs* $(\bar{b}_1,\cdots,\bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq \gamma_2^{n-1}\cdot\det(\Lambda)^{2/n}$, i.e., an $2^{O(n)}$-approximate solution to SVP.
- *Proof.* We prove termination and correctness, in that order. But first we prove a useful lemma that bounds the output of Lagrange reduction in terms of γ_2.
  - **Lemma 1.** $\bar{B}_{i:i+1} := (\pi_i(\bar{b}_i),\pi_i(\bar{b}_{i+1}))$ is Lagrange-reduced iff

    $$\|\bar{b}_i^*\| \leq \gamma_2\cdot\|\bar{b}_{i+1}^*\|.$$

  - Proof.
    - ($\Rightarrow$) If $\bar{B}_{i:i+1}$ is Lagrange-reduced then $\pi_i(\bar{b}_i)=\bar{b}_i^*$ is "locally" shortest for $\bar{B}_{i,i+1}$, and we have

      $$\|\bar{b}_i^*\|^2 \leq \gamma_2\cdot\det(\bar{B}_{i:i+1}) = \gamma_2\|\bar{b}_i^*\|\|\bar{b}_{i+1}^*\|,$$

      where the inequality follows by definition of γ_2, while the equality is a consequence of Homework 1. Dividing the above by $\|\bar{b}_i^*\|$ proves the if part.
    - ($\Leftarrow$) The iff part can be proved by reversing the above steps.

1. *Termination.* We proceed by induction in dimension n: assuming that Algorithm 2 terminates for lattices of dimension *at most* n, we show that Algorithm 2 terminates for lattices of dimension n. (Note that the base case is the same as Lagrange's algorithm.)
   - First note that since the application of Lagrange reduction is "local", $\bar{b}_1$'s norm doesn't change if its is *not* picked for Lagrange reduction.
   - When $\bar{b}_1$ *is* picked for Lagrange reduction, it can be argued using Lemma 1 that its norm after the reduction is *strictly reduced*. To see why, note that Hermite reduction applies Lagrange reduction to $\bar{B}_{1:2} := (\pi_1(\bar{b}_1)=\bar{b}_1,\pi_1(\bar{b}_2))$ only when the basis is *not* Lagrange reduced. Thus *before* the reduction, by Lemma 1, we have

     $$\|\bar{b}_1\| = \|\bar{b}_1^*\| > \gamma_2\cdot\bar{b}_{i+1}^* \Leftrightarrow \|\bar{b}_1^*\|^2 > \gamma_2\cdot\det(\Lambda_{1:2}).$$

     But *after* the reduction we have

$$\|\bar{b}_1\|^2 \leq \gamma_2 \cdot \det(\Lambda_{1:2}),$$

and thus the norm has decreased.

- Since λ_1 is always achieved (i.e., infimum is minimum), the algorithm must halt at some point. After this point, the algorithm works only over $\Lambda_{2:n}$, and hence terminates by induction hypothesis. Thus Algorithm 2 terminates also for dimension n.

2. *Correctness.* Since Algorithm 2 terminates, it must be the case that each pair $(\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))$ is Lagrange reduced and thus (by Lemma 1) satisfies

$$\|\bar{b}_i^*\| \leq \gamma_2 \cdot \|\bar{b}_{i+1}^*\|.$$

It follows that

$$\|\bar{b}_1\| = \|\bar{b}_1^*\| \leq \gamma_2^{i-1} \|\bar{b}_i^*\|.$$

Multiplying the above equation for $i = 1, \cdots, n$, we get

$$\|\bar{b}_1\|^n \leq \Pi_{i=1}^n \gamma_2^{i-1} \|\bar{b}_i^*\|^2 = \gamma_2^{n(n-1)/2} \det(\Lambda),$$

where the last equality follows by Homework 1. Correctness now follows by taking nth root on both sides.

- **Homework 2.** Show that correctness of Algorithm 2 implies the second part of Theorem 2 (Hermite's bound).

- Weak LLL Reduction
  - Note that we are only able to prove that $\|\bar{b}_1\|$ reduces in each iteration, but there is no control over the amount by which the norm reduces, and thus it is unclear whether Algorithm 2 is poly-time. To resolve this, we introduce a "slack" ε in Algorithm 2, called the Lovàsz condition. That is, the algorithm only spends time on doing a Lagrange reduction whenever $\|\bar{b}_i\|$ is *significantly* larger than what it could be. Introducing this slack allows us to show that in each iteration the norm must reduce by a factor dependent on ε and argue polynomial number of iterations.
  - **Definition 4.** Let $\bar{B}$ be a lattice basis. For $i \in [n-1]$ and a parameter $\varepsilon > 0$, $(\bar{b}_i, \bar{b}_{i+1})$ satisfies ε-Lovàsz condition if

$$\|\bar{b}_i^*\| \leq (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|.$$

  - **Algorithm 3**:
    - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\epsilon > 0$
    - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \epsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
    - Algorithm:
      1. While $\exists i \in [n]$ such that $(\bar{b}_i, \bar{b}_{i+1})$ is not ε-Lovàsz reduced, do:
         1. Find unimodular $\bar{U} \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U} = (\pi_i(\bar{b}_i), \pi_i(\bar{b}_{i+1}))\bar{U}$ is Lagrange reduced

    2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}'_i, \bar{b}'_{i+1}, \bar{b}_{i+2}, \cdots, \bar{b}_n)$
  2. Output $\bar{B}$.

- **Theorem 3.** Algorithm 3 *terminates* after $poly(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon))$ iterations and *outputs* $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$ (i.e., an $2^{O(n)}$-approximate solution to SVP).
- *Proof.* Correctness and termination can be argued as for Algorithm 2. Below we focus on bounding the number of iterations. To this end, we use a potential argument. Define

$$P(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i}) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1}$$

  to be potential of a basis $\bar{B}$. Below we will show that $P(\bar{B})$ has a lower bound and that $P(\bar{B})$ decreases by a constant factor in each iteration of Algorithm 3.
  1. To see why the latter is true, observe that $\Lambda_{1:i} \subseteq \mathbb{Z}^n$ and thus $\det(\Lambda_{1:i})$ is an integer and positive thanks to the expression above
  2. First observe that an application of Lagrange reduction on $\bar{B}_{i:i+1}$ only changes $\det(\Lambda_{1:i})$ and leaves the rest of the determinants in the product unchanged.
     - **Homework 3.** Prove the above observation.
  3. Before this application of Lagrange reduction, we have

$$\|\bar{b}_i^*\|^2 > (\gamma_2 + \varepsilon) \cdot \det(\Lambda_{i:i+1}),$$

     and after it we have

$$\|\bar{b}_i^*\|^2 \leq \gamma_2 \cdot \det(\Lambda_{i:i+1}).$$

     Thus the potential reduces by a factor $\rho := \sqrt{\gamma_1/(\gamma_2 + \varepsilon)}$ during this iteration. Note that the initial potential is bounded by

$$\Pi_{i=1}^n \|\bar{b}_i\| = \|\bar{B}\|_\infty^{n(n+1)/2},$$

     where

$$\|\bar{B}\|_\infty := \max_{1 \leq i \leq n} \|\bar{b}_i\|.$$

     Thus the number of iterations N is such that $\|\bar{B}\|_\infty^{n(n+1)/2} \cdot \rho^N \geq 1$, and thus

$$(n(n+1)/2) \log(\|\bar{B}\|_\infty) + N \log(\rho) \geq 0.$$

     Thus $N \leq (n(n+1)) \log(\|\bar{B}\|_\infty) / -2\log(\rho)$. The claim follows by observing that

$$-\log(\rho) = \log(1 + \varepsilon/\gamma_2)/2 = \varepsilon/2\gamma_2 + O(\varepsilon^2).$$

Further Reading

- You can read the recursive variant of Hermite's algorithm in [3], Chapter 2: see Algorithm 5.

2026s-CS758-Lecture07

Topics

- LLL algorithm
- Size-reduction

Sources

1. Intro to Lattice Algorithms and Cryptography, Lecture 4
2. Intro to Lattice Algorithms and Cryptography, Lecture 2
3. Lattices Algorithms and Applications, Lecture 04
4. Lattices Mini Course, Lecture 05

Plan

- Recall/Plan
  - In the last lecture:
    - We saw the *weak* Lenstra-Lenstra-Lovàsz (WLLL) reduction, which solves SVP up to an *exponential* approximation factor. The algorithm is essentially a relaxation of Hermite's reduction in which the Lagrange reduction is done only whenever $\|\bar{b}_i\|$ is *significantly larger* than what it could be, where "significantly larger" is determined by a parameter ε. Recall that Hermite's algorithm itself carries out Lagrange reduction "locally".
    - **Algorithm 1 (WLLL reduction).**
      - *Input*: Basis $(\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\varepsilon > 0$
      - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
      - Algorithm:
        1. While $\exists i \in [n]$ such that $\|\bar{b}_i^*\| > (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|$, do:
           1. Find unimodular $\bar{U}_L \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U}_L$ is Lagrange reduced
           2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}_i', \bar{b}_{i+1}', \bar{b}_{i+2}, \cdots, \bar{b}_n)$
        2. Output $\bar{B}$.
    - We then argued that WLLL reduction runs for at most a polynomial number of *iterations*. However a polynomial run-*time* is not necessarily guaranteed since an iteration itself might run for super-polynomial time. This happens if:
      1. Any entry in $\bar{B}^*$, which is composed of rational numbers, has large numerator or denominator; or
      2. Any entry in $\bar{B}$, which is composed of integers, is large.
  - In today's lecture:

    - We will resolve the above issues today. We will see that the first is already not an issue in WLLL. Then we will see the (full) LLL reduction, which additionally carries out a "size-reduction" step in order to keep the size of entries of $\bar{B}$ in check.
    - We'll describe LLL reduction using the size-reduction algorithm as a black-box. Then in the next lecture we'll see how to size-reduce using Babai's nearest hyperplane algorithm.
- LLL Reduction
    - Recall that the second condition for an Lagrange-reduced basis (Lecture 05, Definition 1) bounds the Gram-Schmidt coefficient, i.e.,

      $$|\mu_{1,2}| \leq 1/2,$$

      where

      $$\mu_{1,2} := \langle \bar{b}_1^*, \bar{b}_2 \rangle / \|\bar{b}_i^*\|^2.$$

      The notion of size-reduction is essentially a *generalisation* of this condition to higher dimensions (we will see a geometric interpretation of this later). In Lemma 1, we will see why this notion is useful: it guarantees that the norm of each basis vector and its Gram-Schmidt counterpart is "not too large" and this, in turn, implies that their numerator is also "not too large" (show this). In fact everything can be represented using $\mathrm{poly}(n)$ bits.
    - **Definition 1 ([1], Definition 12).** A basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice Λ is said to be *size-reduced* if for all $1 \leq i < j \leq n$,

      $$|\mu_{i,j}| \leq 1/2,$$

      where $\mu_{i,j} := \langle \bar{b}_i^*, \bar{b}_j \rangle / \|\bar{b}_i^*\|^2$.
    - **Lemma 1 ([1], Lemma 14).** Recall the definition of potential function of a lattice $\Lambda(\bar{B}) \subset \mathbb{Z}^n$ from Lecture 06:

      $$\Phi(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i} = \Lambda(\bar{B}_{1:i})) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1.} \leq \det(\Lambda)^n$$

      If $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ is size-reduced, then for all $i \in [n]$,

      $$\log(\|\bar{b}_i\|), \log(\|\bar{b}_i^*\|) \leq \mathrm{poly}(n) \cdot \log(\Phi(\bar{B})) \leq \mathrm{poly}(n) \cdot \log(\det(\Lambda)).$$

    - *Proof.* We first prove the second bound. Since $\bar{B} \subset \mathbb{Z}^n$, for all $1 \leq i \leq n$, $\det(\Lambda_{1:i} = \Lambda(\bar{B}_{1:i})) \geq 1$ (as well an integer) and thus $\det(\Lambda_{1:i}) \leq \Phi(\bar{B})$. Since

      $$\|\bar{b}_i^*\| = \det(\Lambda_{1:i}) / \det(\Lambda_{1:i-1}) \leq \det(\Lambda_{1:i}) \leq \Phi(\bar{B}),$$

      taking log on both proves the second bound. Next, since $\bar{B}$ is size-reduced, we can write any basis vector $\bar{b}_i$ as

      $$\bar{b}_i := \bar{b}_i^* + \sum_{j=1}^{i-1} x_i \bar{b}_j^*$$

for some $-1/2 \leq x_i < 1/2$. It follows that

$$\|\bar{b}_i\| \leq \|\bar{b}_i^*\|^2 + \sum_{j=1}^{i-1} |x_i| \|\bar{b}_j^*\|^2 \leq n \cdot \Phi(\bar{B}).$$

Taking log on both sides proves the first bound.

- LLL reduction keeps the entries of the basis vectors and its Gram-Schmidt counterparts in check by additionally (to WLLL reduction) carrying out size-reduction.
  - **Definition 2 ([1], Definition 13).** A basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of lattice Λ is said to be *ε-LLL-reduced* if it is
    1. Size reduced
    2. ε-WLLL reduced, i.e., for every $1 \leq i < n$, $(\bar{b}_i, \bar{b}_{i+1})$ satisfies the ε-Lovàsz condition, i.e.,

$$\|\bar{b}_i^*\| \leq (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|.$$

- Assuming that a basis can be efficiently size-reduced, we now describe the LLL reduction (with difference from WLLL reduction highlighted). In the theorem that follows, we analyse it.
- **Algorithm 2 (LLL reduction).**
  - *Input*: Basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_n)$ of a lattice $\Lambda \subseteq \mathbb{Z}^n$ and a parameter $\varepsilon > 0$
  - *Output*: A basis $(\bar{b}_1, \cdots, \bar{b}_n)$ that is ε-LLL-reduced and thus with $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$
  - Algorithm:
    0. Size-reduce $\bar{B}$
    1. While $\exists i \in [n]$ such that $\|\bar{b}_i^*\| > (\gamma_2 + \varepsilon) \cdot \|\bar{b}_{i+1}^*\|$., do:
       1. Find unimodular $\bar{U}_L \in \mathbb{Z}^{2\times 2}$ such that $(\bar{b}_i', \bar{b}_{i+1}') := \bar{B}_{i:i+1}\bar{U}_L$ is Lagrange reduced
       2. Update $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_{i-1}, \bar{b}_i', \bar{b}_{i+1}', \bar{b}_{i+2}, \cdots, \bar{b}_n)$
       3. Size-reduce $\bar{B}$
    2. Output $\bar{B}$.
- **Theorem 2.** Algorithm 2 *outputs* $(\bar{b}_1, \cdots, \bar{b}_n)$ such that $\|\bar{b}_1\|^2 \leq (\gamma_2 + \varepsilon)^{n-1} \cdot \det(\Lambda)^{2/n}$ in time $T_{LLL} := \mathrm{poly}(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon)$.
- *Proof.* Correctness and termination in $\mathrm{poly}(n, \log(\|\bar{B}\|_\infty, 1/\varepsilon)$ iterations can be argued as in WLLL once you show that size-reduction doesn't affect the potential as we defined in Lecture 06.
  - **Homework 2.** Show that size-reduction does not affect the potential

$$P(\bar{B}) := \Pi_{i=1}^n \det(\Lambda_{1:i}) = \Pi_{i=1}^n \|\bar{b}_i^*\|^{n-i+1},$$

    by showing that the Gram-Schmidt basis is unchanged.
- What remains to show is that the entries of $\bar{B}$ and $\bar{B}^*$ are bounded at all times, and thus each iteration runs in poly-time. To this end, we need to bound the numerator

and denominator of their entries. In the first item below, we bound the denominators of Gram-Schmidt basis $\bar{B}^*$ in terms of the determinant of $\bar{B}$. Note that we have already bounded the norm of size-reduced basis and its Gram-Schmidt counterpart, and thus we are indirectly bounding the numerator. What remains is to show that Lagrange reducing a basis does not blow up the entries: this we prove in the second item below.

1. We first bound the denominators of the Gram-Schmidt orthogonal basis $\bar{B}^*$. Note that this is a general claim, and applies also to WLLL reduction.
   - **Claim 1 ([1], Lemma 11).** If $\bar{B} \in \mathbb{Z}^{n \times n}$ then
   $$\bar{B}^* \in \frac{1}{\det(\bar{B}^t \bar{B})} \mathbb{Z}^{n \times n}.$$
   - *Proof.* We proceed by induction over the dimension n. Consider the projected basis
   $$\bar{B}_{2:n} = (\pi_2(\bar{b}_2), \cdots, \pi_2(\bar{b}_n)).$$
   First observe that the scaled basis
   $$\|\bar{b}_1\|^2 \cdot \bar{B}_{2:n} \in \mathbb{Z}^{n \times (n-1)}$$
   since, by definition, for any $i \in [2, n]$
   $$\pi_2(\bar{b}_i) := \bar{b}_i - \langle \bar{b}_i, \bar{b}_1 \rangle / \|\bar{b}_1\|^2.$$
   On applying the induction hypothesis to $\|\bar{b}_1\|^2 \cdot \bar{B}_{2:n}$, we get
   $$(\|\bar{b}_1\| \cdot \bar{B}_{2:n})^* = \|\bar{b}_1\| \cdot \bar{B}^*_{2:n} \in \frac{1}{\|\bar{b}_1\|^2 \cdot \det(\bar{B}^t_{2:n} \bar{B}_{2:n})} \mathbb{Z}^{n \times (n-1)}.$$
   Since $\bar{B}_{2:n}$ is full rank,
   $$\det(\bar{B}_{2:n} \bar{B}^t_{2:n}) = \Pi_{i=2}^n \|\bar{b}^*_i\|^2.$$
   Since $\bar{b}^*_1 = \bar{b}_1$, we have $\det(\bar{B}\bar{B}^t) = \|\bar{b}_1\|^2 \cdot \det(\bar{B}_{2:n} \bar{B}^t_{2:n})$ and thus
   $$\bar{B}^*_{2:n} \in \frac{1}{\det(\bar{B}\bar{B}^t)} \mathbb{Z}^{n \times n}.$$
   Since each entry $b_{1,j}$ in $\bar{b}_1$ can be expressed as $(b_{1,j} \det(\bar{B}\bar{B}^t)) / \det(\bar{B}\bar{B}^t)$, the claim follows. (Alternatively, $\bar{b}_1$ is orthogonal to each basis vector in $\bar{B}_{2:n}$ and thus $\bar{b}_1 \in 1/\det(\bar{B}\bar{B}^t)\mathbb{Z}^{n \times n}$.)
2. We now prove, assuming Lemma 1, that Lagrange reduction does not blow up the norm of the basis vectors, and hence the size of its entries.
   - **Claim 2 ([1], Lemma 16).** Let $(\bar{b}_1, \bar{b}_2)$ be two linearly-independent vectors, and let $(\bar{b}'_1, \bar{b}'_2) := (\bar{b}_1, \bar{b}_2) \cdot \bar{U}_L$ denote the corresponding Lagrange-reduced basis. Then
   $$\log(\|\bar{b}'_1\|), \log(\|\bar{b}'_2\|) \leq \mathrm{poly}(\log(\|\bar{b}_1\|), \log(\|\bar{b}_2\|)).$$

Moreover, during the Lagrange reduction, all the intermediate basis vectors will also satisfy the above bound.

- *Proof.* We prove the upper bound for $\log(\|\bar{b}_1'\|)$: the proof for $\log(\|\bar{b}_2'\|)$ is similar. Recall from Lecture 05, Theorem 2 that Lagrange reduction on a basis $(\bar{b}_1, \bar{b}_2)$ runs for at most $N_L := O(\log(\|\bar{b}_1\|/\sqrt{\det(\Lambda)})$ iterations. In its ith iteration, the current basis $(\bar{b}_1, \bar{b}_2)$ is multiplied with a unimodular matrix of the form

$$\bar{U}_i := \begin{bmatrix} 0 & 1 \\ 1 & -\lfloor \mu_{1,2} \rceil \end{bmatrix},$$

where $\mu_{1,2} := \langle \bar{b}_1, \bar{b}_2 \rangle / \|\bar{b}_1\|^2$ is the Gram-Schmidt coefficient. By Cauchy-Schwartz inequality, $|\langle \bar{b}_1, \bar{b}_2 \rangle| \leq \|\bar{b}_1\| \|\bar{b}_2\|$ and thus

$$|\lfloor \mu_{1,2} \rceil| \leq \|\bar{b}_2\| / \|\bar{b}_1\|.$$

Moreover, the ℓ_2-matrix norm of $\bar{U}_i$, which is defined as

$$\|\bar{U}_i\| := \max_{\bar{x} \neq 0} \frac{\|\bar{U}_i \bar{x}\|}{\|\bar{x}\|}$$

can be shown to be bounded by

$$\sqrt{\lfloor \mu_{1,2} \rceil^2 + 2} \leq |\lfloor \mu_{1,2} \rceil| + 2.$$

It follows that

$$\|\bar{b}_1'\| \leq \|\bar{b}_1 \bar{U}_L\| = \|\bar{b}_1 \bar{U}_1 \cdots \bar{U}_{N_L}\| \leq \|\bar{b}_1\| \|\bar{U}_1\| \cdots \|\bar{U}_{N_L}\|,$$

which after taking logarithms on both sides, yields

$$\log(\|\bar{b}_1'\|) \leq \log(\|\bar{b}_1\|) + \sum_{i=1}^{N_L} \log(\|\bar{U}_i\|) \leq \log(\|\bar{b}_1\|) + N_L \log(\|\bar{b}_2\| / \|\bar{b}_1\| + 2),$$

which proves the claim.

- ==Size-Reduction ==
  - What remain to answer is how to size-reduce a basis. To this end, we are going to use Babai's Nearest Plane algorithm, which is a close variant of Gram-Schmidt algorithm. We discuss this in Lecture 08. For explaining the BNP algorithm it is instructive to have a geometric interpretation of size-reduction, via Babai's Fundamental Region.
  - Babai's Fundamental Region
    - Recall fundamental parallelepiped of a basis $\bar{B}$ from Lecture 03 (e.g., see $\Lambda((3,1)^t, (1,2)^t)$ in Figure 1).
    - **Definition 2.** Given a basis $\bar{B} := (\bar{b}_1, \cdots, \bar{b}_m) \in \mathbb{R}^{n \times m}$, the *fundamental parallelepiped* associated with $\bar{B}$ is the region

$$P(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : 0 \leq x_i < 1 \right\}$$

- Recall that the (linear) span of $\Lambda(\bar{B})$ can be "tiled" using $P(\bar{B})$: i.e., it is both a covering and a packing of the span (see formal definition below). In particular, when $m = n$, which we will focus on for the rest of the lecture, the ambient space $\mathbb{R}^n$ can be tiled using $P(\bar{B})$. As a result, as you had to prove in Assignment 1, for each point $\bar{z} \in \mathbb{R}^n$, there exists a *unique* "representative" lattice point $\bar{y}$ such that

$$\bar{z} \in \bar{y} + P(\bar{B}).$$

  - **Example 1.** For $\Lambda((3,1)^t, (1,2)^t)$ from Figure 1, the representative of $(5,5)^t$ is the lattice point $((3,1)^t + 2(1,2)^t = (5,5)^t$.
- More generally, one can talk about fundamental regions, defined below.
  - **Definition 3 ([3]).** Let $\Lambda \subset \mathbb{R}^n$ be a full-rank lattice and let $\mathcal{S}$ be a (measurable) set. We define $\mathcal{S}$ to be
    1. Λ-*packing* if $\forall \bar{y}_1 \neq \bar{y}_2 \in \Lambda$, $(\bar{y}_1 + \mathcal{S}) \cap (\bar{y}_2 + \mathcal{S})) = \emptyset$.
    2. Λ-*covering* if $\Lambda + \mathcal{S} = \mathbb{R}^n$.
    3. Λ-*tiling* or a *fundamental region* of Λ if $\mathcal{S}$ is both Λ-packing and Λ-covering.
- A fundamental region related to $P(\bar{B})$ is the *centred half-open parallelepiped* $P'(\bar{B})$, obtained by "shifting" $P(\bar{B})$ (see Figure 1):

$$P'(\bar{B}) := \left\{ \sum_{i=1}^{m} x_i \bar{b}_i : -1/2 \leq x_i < 1/2 \right\}.$$

  The size-reduction requirement can be seen through a fundamental region called the *orthogonalised centred parallelepiped* $P^*(\bar{B})$, a/k/a *Babai's fundamental region*, which is defined in terms of the Gram-Schmidt basis (see Figure 2 for Babai's fundamental region for $\Lambda((3,1)^t, (1,2)^t)$:

$$P^*(\bar{B}) := P'(\bar{B}^*) = \left\{ \sum_{i=1}^{m} x_i \bar{b}_i^* : -1/2 \leq x_i < 1/2 \right\}.$$

- The size-reduction condition can then be geometrically interpreted as requiring the *orthogonal projections* of $\bar{b}_i$ on to $\bar{b}_1^*, \cdots, \bar{b}_{i-1}^*$ (viz, $\bar{b}_i - \bar{b}_1^*, \cdots, \bar{b}_i - \bar{b}_{i-1}^*$) *to lie in* $P^*(\bar{B})$. As noted above this is a generalisation of the second wristwatch condition.
  - **Example 2.** The lattice $\Lambda((3,1)^t, (1,2)^t)$ from Figures 1 and 2 is already size-reduced since $\mu_{1,2} = 5/10 = 1/2$.